ACS 597: Noise Control Applications
Sound sources

1 Simulation of sources from monopoles

One benefit of studying monopole sources is that we can use them to build up sources of arbitrary sizes and shapes.

The provided MATLAB function, `sum_of_monopoles`, calculates the acoustic pressure at a set of receiver locations from a sum of monopoles with specified locations and source strengths. The function can be called with the syntax,

```
p = sum_of_monopoles(xyz_sources, Q_sources, xyz_receivers, f, rho0, c)
```

where `xyz_sources` is a matrix of (x,y,z) source locations, `Q_sources` is a vector of complex source strengths, `xyz_receivers` is a matrix of receiver locations, `f` is a single frequency, `rho0` is the fluid density, and `c` is the speed of sound. The output `p` is the complex pressure amplitude at the receiver locations.

Look through the `sum_of_monopoles` code, and make sure you understand what the various lines of code are doing. You can type `help sum_of_monopoles` in the command line to read the help text on this function and see the size of the input variables needed.

Use the code to help you make the following plots. Where constants are not specified, feel free to make your own choices while making sure to follow the assumptions for the specified source. For example, the two monopoles making up a dipole source need to be much closer than one wavelength apart.

There are two different types of plot you will need to make:

- **Distance dependence**—Put receivers along one axis, starting near the center of the source and extending several wavelengths. Do not put a receiver *exactly* where a monopole is located, since it will have infinite pressure. Plot the magnitude of the pressure against distance from the center of the source, usually on a log-log scale.

- **Directivity**—Put receivers in a circle around the source, several wavelengths away from the source. Plot the pressure amplitude against the angle of the receiver location. You can use the `polar` plot in MATLAB to make the plot in polar coordinates.

(a) Show the spherical spreading from a monopole. Show that the pressure drop with distance is proportional to $\frac{1}{r}$.

(b) Compute and plot the directivity patterns for a monopole, dipole, lateral quadrupole, and longitudinal quadrupole. Show them all in one plot to compare them.

(c) Plot the directivity pattern for a finite line source. Compare several different source arrangements: with sources spaced far apart and close together and extending from less than one wavelength to several wavelengths.

(d) Plot the axial dependence of a baffled piston. The source locations for this simulation can be challenging. Ideally, we want each source to account for an equal area of the piston. A reasonable arrangement for N points inside a circle of radius a is to put them at coordinates[1]

$$(x_n, y_n) = (r_n \cos\theta_n, r_n \sin\theta_n)$$

where

$$\theta_n = \frac{2\pi}{\phi^2} n$$

$$r_n = a\sqrt{\frac{n}{N}}$$

and $n = 0, 1, \ldots, N-1$ and $\phi = \frac{1+\sqrt{5}}{2}$. Note that if the piston is in the x-y plane, the receiver points should be along the z-axis. The receiver points should start close to the piston ($z < a/10$) and extend several wavelengths away. Choose a and f such that a/λ is between 5 and 10 to make sure there are some nulls in the result. Compare the location of the nulls in this simulation to the theoretical location of the nulls,

$$r_{\text{null}} = \frac{(a/\lambda)^2 - m^2}{2m/\lambda}$$

where m is an integer $1 \leq m \leq a/\lambda$.

[1] See https://demonstrations.wolfram.com/SunflowerSeedArrangements/

2 Active control of enclosure opening

A square opening of 200 mm on a side is required for access into an enclosure containing a source of low-frequency noise. Suppose that speakers could be mounted on either side of the opening with centers 205 mm from the center of the opening, and that an actice noise control system could sample the emitted noise in sufficient time before it reaches the opening to provide a cancelling signal to drive the speakers 180° out of phase and at half the ampitude of the sound being emitted from the opening at all frequencies in octave bands between 63 Hz and 500 Hz. The two speakers and the opening would form a longitudinal quadrupole in the wall of the enclosure.

(a) The ratio of the sound power of a monopole source to a longitudinal quadrupole with center source equal to the monopole is

$$\frac{W_{\text{long}}}{W_{\text{mono}}} = \frac{4}{5}(kd)^4,$$

where d is the distance between the center monopole and the side monopoles. What are the sound power level reductions, in dB, at the four octave band center frequencies 63, 125, 250, and 500 Hz?

(b) The ratio of mean square pressures of a longitudinal quadrupole to a monopole is

$$\frac{\overline{p^2}_{\text{long}}}{\overline{p^2}_{\text{mono}}} = 5\cos^4\theta \frac{W_{\text{long}}}{W_{\text{mono}}},$$

where θ is measured from the line connecting the monopole sub-sources. Plot the predicted sound pressure level reductions in dB at angles of $\theta = 0$ to $90°$ for the four center frequencies. Comment on the effectiveness of the active noise control in this situation.

3 Coherent and incoherent speakers

Three speakers are mounted in line and flush with a rigid concrete floor out in the open, away from any other reflecting surfaces. All three speakers are very small compared with the wavelength of sound they radiate. The distance between centers of adjacent speakers is 0.1 m. Calculate the expected sound pressure level at a height of 1 m avoce the floor and a horizontal distance of 20 m from the center speaker at an azimuthal angle of 30° from the line connecting the three speakers for the following two cases:

(a) A 125 Hz pure tone from the same source is fed to all three speakers such that the center speaker is 180° out of phase with the two outer speakers (which are in phase). The center speaker by itself radiates 1 W and the two outer speakers each radiate 0.5 W of sound.

(b) All three speakers are fed 125 Hz one-third octave band random noise at the same power level as the previous part.

4 Traffic noise modeling

Assume a line of traffic on a straight highway can be modeled as an incoherent line source with an average separation distance of 8 m between sources and an average A-weighted source pressure level of 86 dB at a distance of 1 m.

(a) What will the sound pressure level be at house 200 m from the road?

(b) What would the sound pressure level be if instead the highway were a circle of radius 200 m with the house at its center?